Band-1 Onsager Critical Exponent Remaining Endpoint — Lit Probe (Collapse 2.10)

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Honesty stamp. This note **locks exact literature sentences** for Onsager's energy-threshold conjecture on 3D incompressible Euler. It does **not** construct new weak solutions, does **not** claim the endpoint closed, and does **not** stamp Clay NS / ToE / GATE-04. **Not a full close. GATE-04 locked. No Clay / ToE stamp.**

> **Exact sentence lock.** Onsager's conjecture for 3D incompressible Euler is settled for every Hölder exponent strictly below 1/3 (existence of energy-nonconserving weak solutions in $C^{1/3-}$) and for every exponent strictly above 1/3 (energy conservation); the remaining endpoint — existence of energy-nonconserving weak solutions in the exact class $L_t^\infty C_x^{1/3}$ (or $C_t C_x^{1/3}$) — remains **OPEN** as of 2026-09-11.

Abstract

Closed (2017–2018 core + refinements): Isett (*Annals* 2018) constructs energy-nonconserving weak Euler solutions in $C_{t,x}^{\alpha}$ for every $\alpha < 1/3$; Buckmaster–De Lellis–Székelyhidi–Vicol (*CPAM* 2019) realize arbitrary smooth positive energy profiles in the same strict subcritical range. Positive side: conservation for $\alpha > 1/3$ (Eyink; Constantin–E–Titi 1994) and in the Besov class $L_t^3 B_{3,c_0}^{1/3}$ (Cheskidov–Constantin–Friedlander–Shnirelman 2008).

Remaining: exact endpoint $\alpha = 1/3$ nonconservation. Isett (*Analysis & PDE* 17 (2024), 2123–2159) approaches the endpoint with scale-dependent regularity covering $[0, 1/3)$ and studies intermittency constraints; it **does not** close $\alpha = 1/3$.

Separations (mandatory): Buckmaster–Vicol (*Annals* 2019) weak **Navier–Stokes** non-uniqueness is a **different proved theorem**, not an Onsager-endpoint close. Classical $C^{1,\alpha}$ Euler blowup (Elgindi 2021; 2026 axisymmetric threshold papers) is a **different claim surface** (singularity formation, not anomalous dissipation of weak Hölder flows).

Problem (Band-1 collapse item H-2.10)

1. Cite settled vs remaining Onsager status without invention (2017–2026 web/arXiv check).
2. Lock one exact house sentence for $C^{1/3-}$ vs endpoint.
3. Separate Isett/BDLSV Onsager from Buckmaster–Vicol NS non-uniqueness and from classical $C^{1,\alpha}$ blowup.
4. Keep GATE-04, Clay rhetoric, and full-close claims out of the claim surface.

Literature ledger (2017–2026 check; not re-proved on box)

Year	Result	Authors	Status for H-2.10
1994	conservation for $\alpha > 1/3$	Eyink; Constantin–E–Titi	**CLOSED** (positive)
2008	conservation in $L^3_t B^{1/3}_{3,c_0}$	CCFS	**CLOSED** (sharper positive); leaves room for endpoint dissipation
2017–18	nonconservation for all $\alpha < 1/3$ in C^α	Isett, *Annals* 188 (2018)	**CLOSED** (negative subcritical)
2019	flexible energy profiles, all $\beta < 1/3$	Buckmaster–De Lellis–Székelyhidi–Vicol, *CPAM*	**CLOSED** (still strict $< 1/3$)
2019	weak NS non-uniqueness	Buckmaster–Vicol, *Annals* 189	**SIBLING** — not Onsager endpoint
2021	$C^{1,\alpha}$ Euler singularity (axisym.)	Elgindi, *Annals*	**SIBLING** — classical blowup Onsager
2024	approach to endpoint; intermittency	Isett, *APDE* 17 (2024)	**increment**; endpoint still **OPEN**
2025–26	$C^{1,\alpha}$ blowup near 1/3 threshold	various arXiv	**heat** for classical regularity; **not** elevated as Onsager close

Primary citations (settled negative). Philip Isett, *A proof of Onsager's conjecture*, Ann. of Math. **188** (2018), 871–963 (arXiv:1608.08301). Tristan Buckmaster, Camillo De Lellis, László Székelyhidi Jr., Vlad Vicol, *Onsager's conjecture for admissible weak solutions*, Comm. Pure Appl. Math. **72** (2019) (arXiv:1701.08678).

Primary citation (endpoint approach, still open). Philip Isett, *On the endpoint regularity in Onsager's conjecture*, Analysis & PDE **17** (2024), 2123–2159 (arXiv:1706.01549). Abstract: conjecture "recently solved… yet the endpoint case remains an interesting open question."

Primary citations (positive). G. L. Eyink, Physica D **78** (1994); P. Constantin, W. E, E. S. Titi, Comm. Math. Phys. **165** (1994); A. Cheskidov et al., Nonlinearity **21** (2008).

Separation: Isett/BDLSV vs Buckmaster–Vicol NS

Theorem	PDE	What is proved	What is **not** proved
Isett 2018 / BDLSV 2019	**Euler** (inviscid)	energy nonconservation for Hölder $< 1/3$	endpoint $= 1/3$; Clay NS
Buckmaster–Vicol 2019	**Navier–Stokes** (viscous)	non-uniqueness of finite-energy **weak** NS	Leray–Hopf uniqueness; classical smooth blowup; Onsager endpoint; Clay

Both use convex integration. Sharing a method does **not** identify the theorems. House rule: cite BV–NS as sibling milestone; **never** rewrite it as "Onsager closed."

Separation: Onsager vs classical $C^{1,\alpha}$ blowup

Onsager asks whether **weak** low-Hölder Euler flows can **dissipate energy** without viscosity. Elgindi / Chen–Hou / 2026 $C^{1,\alpha}$ papers ask whether **classical** (or strong) solutions can **blow up** in finite time. Different regularity goals, different conclusions, different house items (H-2.10 vs H-1.20 / H-4.13). A 2026 preprint

that reaches blowup for all $\alpha \in (0, 1/3)$ at a $C^{1,1/3}$ **structural** threshold is **not** a construction of an $L_t^\infty C_x^{1/3}$ energy-nonconserving weak solution.

Method (box)

1. Web/arXiv ledger for 2017–2026 status (Isett Annals; BDLSV; Isett APDE 2024; BV–NS; positive CET/CCFS).
2. Exact sentence lock in `experiments/band1/onsager_endpoint/STATEMENT_LOCK.md`.
3. Machine-readable `literature_ledger.json` with honesty flags.
4. **No** numerical Euler solver; **no** invented convex-integration certificate.

Results

Literature (cited, not re-proved)

Quantity	House status
Conservation for $\alpha > 1/3$	**CLOSED**
Nonconservation for all $\alpha < 1/3$	**CLOSED** (Isett; BDLSV)
Endpoint $\alpha = 1/3$ nonconservation	**OPEN**
Buckmaster–Vicol weak NS non-uniqueness	**CLOSED** (different theorem)
Classical C^∞ / Clay unforced NS	**OPEN** (out of this probe)

Box

Literature + statement lock only. No computational PDE run.

Closed vs remaining (return surface)

Closed tonight (lit lock):

- Positive Onsager for $\alpha > 1/3$ (+ Besov refinement).
- Negative Onsager for every $\alpha < 1/3$ ($C^{1/3-}$), including flexible energies.
- Precise separation from BV–NS non-uniqueness and from classical $C^{1,\alpha}$ blowup.

Remaining:

- Exact endpoint: energy-nonconserving weak Euler in $L_t^\infty C_x^{1/3}$ / $C_t C_x^{1/3}$.
- Related open (not claimed closed): vanishing-viscosity realization of dissipative Euler; full intermittency program at critical regularity.

Non-claims

- Do **not** say "Onsager solved" without the $C^{1/3-}$ / endpoint nuance.
- Do **not** equate Buckmaster–Vicol 2019 NS non-uniqueness with the Onsager endpoint.
- Do **not** stamp Clay NS, unforced smooth Euler blowup, or finished ToE.
- GATE-04 collection remains **LOCKED**.

Paths

- Lab: `experiments/band1/onsager_endpoint/`
- Paper: `papers/2026-09-11_Band1_Onsager_endpoint_probe.md` (+ PDF)
- Ledger: `experiments/band1/onsager_endpoint/literature_ledger.json`

References (selected)

1. L. Onsager, *Nuovo Cimento* **6** (1949), Suppl. 2, 279–287.
2. G. L. Eyink, *Physica D* **78** (1994), 222–240.
3. P. Constantin, W. E, E. S. Titi, *Comm. Math. Phys.* **165** (1994), 207–209.
4. A. Cheskidov, P. Constantin, S. Friedlander, R. Shvydkoy, *Nonlinearity* **21** (2008), 1233–1252.
5. P. Isett, *Ann. of Math.* **188** (2018), 871–963.
6. T. Buckmaster, C. De Lellis, L. Székelyhidi Jr., V. Vicol, *Comm. Pure Appl. Math.* **72** (2019).
7. P. Isett, *Analysis & PDE* **17** (2024), 2123–2159.
8. T. Buckmaster, V. Vicol, *Ann. of Math.* **189** (2019), 101–144. *(NS non-uniqueness — sibling only.)*